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BTECH
(SEM VI) THEORY EXAMINATION 2023-24
COMPUTER NETWORKS

TIME: 3 HRS

M.MARKS: 100

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2 x 10 = 20

a.	What are the applications of Computer Networks?	02
b.	Write advantages and disadvantages of Mesh Topology.	02
c.	Define fixed and variable size framing.	02
d.	Explain the working of Bridge.	02
e.	Define delivery in Network Layer.	02
f.	Define Token Bucket.	02
g.	Write about Registered ports.	02
h.	Differentiate between connection less and connection oriented.	02
i.	What is Telnet?	02
j.	Define Compression.	02

SECTION B

2. Attempt any three of the following:

3 x 10 = 30

a.	Define TCP/IP Model in detail.	10
b.	Explain Noisy channel Protocols with explanation and diagram.	10
c.	Define IPv4 header Format in detail.	10
d.	Define TCP features and header in detail.	10
e.	Explain SNMP in detail.	10

SECTION C

3. Attempt any one part of the following:

1 x 10 = 10

a.	Name and explain Network Devices on each layer of OSI Model.	10
b.	Define various topologies with diagram, features, and pros-cons.	10

4. Attempt any one part of the following:

1 x 10 = 10

a.	Define any 5 LAN standards.	10
b.	Explain all Random-Access protocols.	10

5. Attempt any one part of the following:

1 x 10 = 10

a.	Explain all Interdomain and Intradomain routings algorithms.	10
b.	Senders' data D= 110101, CRC generator polynomial= x^3+x+1 . Apply CRC algorithm and Perform calculations both at sender and receiver end.	10

6. Attempt any one part of the following:

1 x 10 = 10

a.	Define Connection Management. Explain how Three-way handshaking using diagram.	10
b.	Write about Quality-of-Service Parameters.	10

7. Attempt any one part of the following:

1 x 10 = 10

a.	Differentiate between FTP and HTTP.	10
b.	Explain Cryptography. Write about RSA Algorithm with example.	10